DELA AKPALU

Data Scientist | Statistician | Research Analytics

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Profile

Statistician and data scientist with graduate training from Wake Forest University and AIMS. Builds reproducible Python, R, and SQL workflows spanning causal inference, experimentation, forecasting, survival analysis, and machine learning. Combines clinical research data experience with modular software pipelines, held-out evaluation, and clear technical communication.

Technical Skills

Programming: Python, R, SQL, C++, Julia, Git, LaTeX
Statistical Methods: Probability, statistical inference, regression, Bayesian statistics, causal inference, A/B testing, difference-in-differences, survival analysis, experimental design, stochastic processes
Machine Learning: scikit-learn, PyTorch, classification, anomaly detection, feature engineering, neural networks, uplift modeling, model selection, precision-recall analysis
Time Series and Data Systems: VAR, recurrent neural networks, nowcasting, temporal validation, Spark, MySQL, REDCap, modular Python pipelines, model persistence, CI
Visualization and Delivery: Matplotlib, Tableau, Power BI, technical reporting, dashboard design, Jira

Selected Projects

Public case studies: dela-ek.github.io. Project repositories are private; code available on request.

Causal Inference and Experimentation *Python, randomization, DiD*
Case study
- Analyzed 42,613 real randomized email assignments and 378 paired NJ/PA fast-food interviews; estimated conversion risk differences and employment DiD with balance, attrition, and identification checks.

Clinical Trial Analysis and Missing Outcomes *Python, ANCOVA, HC3*
Case study
- Reanalyzed two ACTG 175 randomized arms (1,054 participants), adjusting 96-week CD4 outcomes for baseline covariates; quantified uncertainty and sensitivity to 38.0% missing follow-up.

ACS Survey Estimation *Python, survey weights, SDR*
Case study
- Used 7,554 actual Delaware adult PUMS records and all 80 official replicate weights; estimated bachelor-degree prevalence of 36.4% with design-based uncertainty and age-domain comparisons.

Retail Sales Forecasting with Uncertainty *Python, temporal validation*
Case study
- Built a modular one-day forecasting pipeline on 1,913 actual M5 sales days; test MAE 193.3 versus seasonal-naive 316.0 units, with 90.5% coverage for nominal 90% calibrated intervals.

Campaign Response Uplift *Python, T-learners, IPW*
Case study
- Selected response models on separate validation data and evaluated a fixed 30% email policy on 8,523 held-out randomized customers; reported IPW incremental conversion and its uncertainty.

Time to Repeat Purchase *Python, Kaplan–Meier, censoring*
Case study
- Analyzed real invoices for 5,878 customers with administrative right censoring, pointwise survival intervals, and restricted mean waiting time; distinguished repeat purchase from unobserved churn.

Additional real-data projects: Card fraud (284,807 transactions; chronological test AP 0.744); activity recognition (91.3% accuracy on nine held-out people); retail recommendations; retrospective macro nowcasting; SQL funnels; segmentation; sensor drift monitoring. All projects include reproducible Python modules, provenance, tests, reports, and CI.

Research and Professional Experience

Analytics and Impact Lead **2026–Present**
Greater Faith Deliverance Ministries / Planning Committee / YAYA Tutoring *Project-Based / Volunteer*
- Develop a KPI and analytics framework for program outcomes, volunteer capacity, and operational progress across 21 auxiliaries.
- Support data collection, dashboarding, STEM pilot evaluation, survey design, engagement measurement, and impact reporting.
- Configure Jira workflows to track initiatives, owners, deadlines, blockers, and quarterly progress.

Research Assistant **Jun 2024–May 2025**
Wake Forest University *Winston-Salem, NC*
- Improved REDCap data quality and validation to support reliable storage, retrieval, and analysis of clinical research datasets.
- Automated data-cleaning and validation workflows in R, reducing manual processing and strengthening reproducibility.
- Collaborated with research teams to structure, maintain, and prepare large datasets for statistical analysis and reporting.

Graduate Teaching Assistant **Aug 2023–May 2025**
Wake Forest University *Winston-Salem, NC*
- Taught and supported statistics, inference, exploratory analysis, visualization, and reproducible R/Python workflows.
- Received the 2025 Department of Statistical Sciences Distinguished Teaching Assistant Award.

Mathematics Teacher **Nov 2025–Present**
St. Vincent Academy *Newark, NJ*
- Teach Algebra II, Geometry, and Calculus, emphasizing mathematical reasoning, quantitative fluency, and clear problem-solving.
- Use assessment data to identify learning gaps and develop targeted LaTeX-based notes, assessments, and worked examples.

Teaching Assistant **Sep 2020–Aug 2022**
Kwame Nkrumah University of Science and Technology *Kumasi, Ghana*
- Led instruction in experimental design, algebra, calculus, and statistical methods; mentored students on research and data projects.

Education

M.S., Statistical Science **Aug 2023–May 2025**
Wake Forest University *Winston-Salem, NC*
Causal Inference, Data Mining, Bayesian Statistics, Machine Learning, Survival Analysis

Structured M.Sc., Data Science **Aug 2022–Jun 2023**
African Institute for Mathematical Sciences *Accra, Ghana*
Big Data Analytics, Network Science, High-Dimensional Data Analysis

B.S., Statistics **Sep 2016–Aug 2020**
Kwame Nkrumah University of Science and Technology *Kumasi, Ghana*
Regression Analysis, Demographic Statistics, Stochastic Processes

Publication, Award, and Certification

- **Coauthor, Frontiers in Aging (2025).** "Bone mineral density and turnover response to GLP-1 receptor agonists in older adults with overweight/obesity and prediabetes/type 2 diabetes: a 20-week pilot trial post hoc analysis."
- **Distinguished Teaching Assistant Award,** Wake Forest University Department of Statistical Sciences, 2025.
- **Data Science Certificate,** Wake Forest University, 2025.

Project results above are educational or research experiments and are not investment, medical, financial, legal, or operational advice.